# LOCALIZING AWAY FROM A FINITE SET OF PRIMES GIVES A SEMI-LOCAL RING WHOSE MAXIMAL IDEALS ARE THE MAXIMAL ELEMENTS OF THE SET

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*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

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*Allowed standard prerequisites (editorial).*

- Spectrum of a localization and avoidance of finitely many primes. Stacks Project, Algebra, Lemma algebra-lemma-spec-localization (tag 00E4) and Lemma algebra-lemma-silly (tag 00EL). Together they describe which primes survive in the localization and hence which become maximal.

*Source reference.* Original reference 00E3: sources/stacks-chow/tags.txt, source lines 520–520. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 00DS: sources/stacks-chow/tags.txt, source lines 508–508. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $A$ be a ring. Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ be a finite set of a primes of $A$. Let $S = A \setminus \bigcup \mathfrak{p}_i$. Then $S$ is a multiplicative system and $S^{-1}A$ is a semi-local ring whose maximal ideals correspond to the maximal elements of the set $\{\mathfrak{p}_i\}$.*

*Proof.* If $a, b \in A$ and $a, b \in S$, then $a, b \notin \mathfrak{p}_i$ hence $ab \notin \mathfrak{p}_i$, hence $ab \in S$. Also $1 \in S$. Thus $S$ is a multiplicative subset of $A$. By the description of $\mathrm{Spec}(S^{-1}A)$ in Algebra, Lemma 00E3 and by Algebra, Lemma 00DS we see that the primes of $S^{-1}A$ correspond to the primes of $A$ contained in one of the $\mathfrak{p}_i$. Hence the maximal ideals of $S^{-1}A$ correspond one-to-one with the maximal (w.r.t. inclusion) elements of the set $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_r\}$. □

*Date*: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.